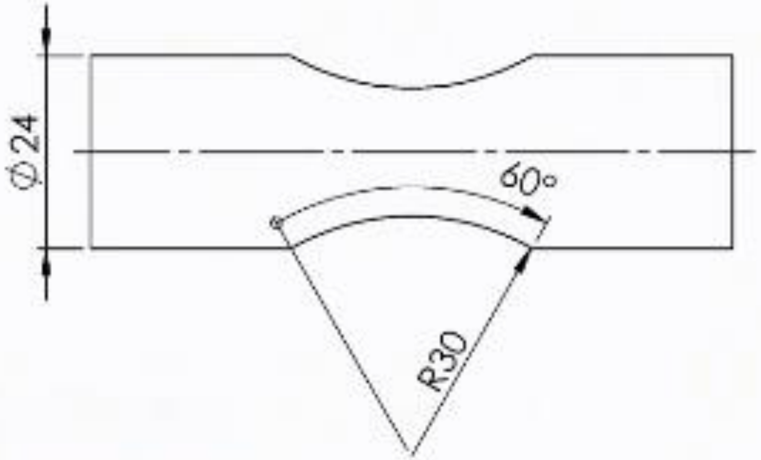
Problem 4 (21/06/2021)

In the figure, the final geometry of a mechanical component is represented. To reach that shape, an external finishing operation is made on a lathe, mounting the workpiece between centers, and considering the following data: $a_p = 1$ mm, $k_{cs} = 2100\,\mathrm{MPa}$, $x = 0{,}29$; $\kappa_r = 65°$, $\kappa_r' = 65°$; component length equal to 90 mm.

1. Considering $n = 3200\,RPM$, determine the maximum and the minimum cutting speeds.
2. Considering $n = 3200\,RPM$ and $f = 0{,}15\,\mathrm{mm/rev.}$, determine the cutting force and power to machine the external cylindrical surface.
3. The company estimates for each machined component a tool contact time of 40 s and a loading / unloading time of 30 s. It is planned to use tool inserts with two cutting edges each, which guarantee (under current cutting conditions) a tool life per cutting edge equal to 4,5 minutes. The tool (cutting edge) changing time is equal to 30 s. The cutting edge cannot be replaced during a component machining, and, at the end of the shift, the last insert mounted on the tool body is discarded even if it has a residual life sufficient for other machining operations or a completely new cutting edge. Calculate the production time per part (T_p), the number of parts that can be machined in an 8-hour shift, and the total cost of tool inserts in an 8-hour shift knowing that the unit cost is 8 €/insert.



MACHINING : TURNING OPERATIONS

- ***Solution - Q1***

Considering $n = 3200\,RPM$, ***determine the maximum and the minimum cutting speeds****.*

The machining operation is done considering a constant revolution per minute; therefore, **the maximum cutting speed** is the cutting speed at **the maximum final diameter**, equal to $D_{max} = 24\,mm$.

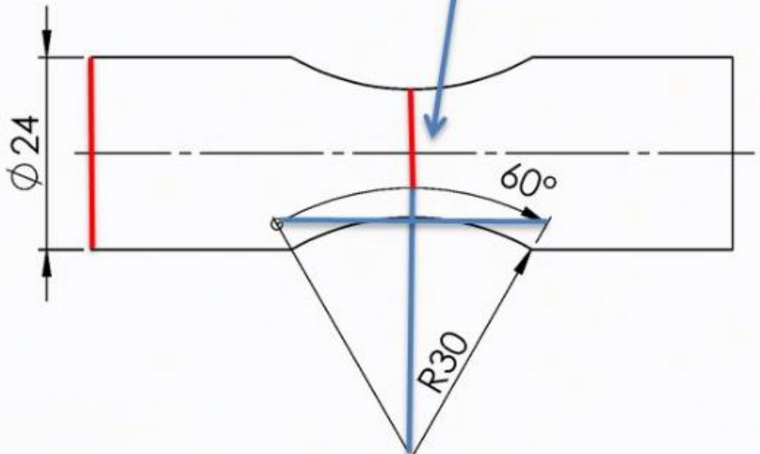
$$v_{c,max} = \frac{n \cdot \pi D_{max}}{1000} = \frac{3200 \cdot \pi \cdot 24}{1000} = 241{,}27 \text{ m/min}$$

Similarly, **the minimum cutting speed** is the cutting speed at **the minimum final diameter**, equal to

$$D_{min} = D_{max} - 2 \cdot R \cdot \left(1 - \cos {}^{\alpha}/_{2}\right) = 24 - 2 \cdot 30 \cdot \left(1 - \cos {}^{60°}/_{2}\right) = 15{,}96 \text{ mm}$$

Therefore,

$$v_{c,min} = \frac{n \cdot \pi D_{min}}{1000} = \frac{3200 \cdot \pi \cdot 15{,}96}{1000} = 160{,}45 \text{ m/min}$$



- ***Solution - Q2***

Considering $n = 3200\,RPM$ *and* $f = 0{,}15\,mm/rev.$, *determine* ***the cutting force and power*** *to machine the external cylindrical surface.*

$$k_c = k_{cs}/{h_D}^x \text{,where } k_{cs} = k_{c0,4} \cdot 0{,}4^x \text{ and } h_D = f \cdot \sin k_r$$

$$F_c = k_c\, A_D = k_c a_p f$$

The cutting force is equal to

$$\boldsymbol{F_c} = \frac{k_{cs}}{(sin\,\kappa_r)^x} \cdot f^{1-x} \cdot a_p = \frac{2100}{(sin\,65°)^{0{,}29}} \cdot 0{,}15^{1-0{,}29} \cdot 1 = 561{,}87 \text{ N}$$

Therefore, the cutting power is equal to

$$\boldsymbol{P_c} = \frac{F_c \cdot v_{c,max}}{60} = \frac{561{,}87 \cdot 241{,}27}{60} = 2.259{,}4 \text{ W} = 2{,}3 \text{ kW}$$

- ***Solution - Q3***

The company estimates for each machined component ***a tool contact time of 40 s*** *and* ***a loading / unloading time of 30 s****. It is planned to use* ***tool inserts with two cutting edges each****, which guarantee (under current cutting conditions)* ***a tool life per cutting edge equal to 4,5 minutes****. The* ***tool*** *(cutting edge)* ***changing time is equal to 30 s****. The cutting edge cannot be replaced during a component machining and, at the end of the shift, the last insert mounted on the tool body is discarded even if it has a residual life sufficient for other machining operations or a completely new cutting edge. Calculate* ***the production time per part*** (T_p)***, the number of parts*** *that can be machined in an 8-hour shift, and* ***the total cost of tool inserts*** in an 8-hour shift *knowing that the unit cost is 8 €/insert.*

The production time per part $(\mathrm{T_p})$ is equal to

$$T_p = T_h + T_m + \frac{T_t}{n_p}$$

where n_p is the number of parts a cutting edge is able to machine, equal to

$$n_p = \left\lfloor \frac{T}{T_m} \right\rfloor = \left\lfloor \frac{4{,}5 \cdot 60}{40} \right\rfloor = \lfloor 6{,}75 \rfloor = 6$$

Therefore,

$$\boldsymbol{T_p} = T_h + T_m + \frac{T_t}{n_p} = 30 + 40 + \frac{30}{6} = 75\ \mathrm{s}$$

The number of parts that can be machined in an 8-hour shift is

$$\boldsymbol{n_{parts}} = \left\lfloor \frac{T_{shift}}{T_p} \right\rfloor = \left\lfloor \frac{8 \cdot 3600}{75} \right\rfloor = 384\ \mathrm{parts/shift}$$

- ***Solution - Q3***

The company estimates for each machined component ***a tool contact time of 40 s*** *and* ***a loading / unloading time of 30 s****. It is planned to use* ***tool inserts with two cutting edges each****, which guarantee (under current cutting conditions)* ***a tool life per cutting edge equal to 4,5 minutes****. The* ***tool*** *(cutting edge)* ***changing time is equal to 30 s****. The cutting edge cannot be replaced during a component machining and, at the end of the shift, the last insert mounted on the tool body is discarded even if it has a residual life sufficient for other machining operations or a completely new cutting edge. Calculate* ***the production time per part*** (T_p), ***the number of parts*** *that can be machined in an 8-hour shift, and* ***the total cost of tool inserts*** in an 8-hour shift *knowing that the unit cost is 8 €/insert.*

The number of cutting edges to be used in a shift production is

$$n_{cutting\ edges} = \left\lceil \frac{n_{parts}}{n_p} \right\rceil = \left\lceil \frac{384}{6} \right\rceil = 64$$

Considering two cutting edges per tool insert, the required number of tool inserts per shift is equal to

$$n_{inserts} = \left\lceil \frac{n_{cutting\ edges}}{2} \right\rceil = \left\lceil \frac{64}{2} \right\rceil = 32$$

Therefore, the total cost of tool inserts is

$$\boldsymbol{C_{inserts}} = n_{inserts} \cdot c_{insert} = 32 \cdot 8 = 256\ €$$